

BHOOMI OS — AGRICULTURAL EVIDENCE LAYER

Automated Sentinel-2 & Meteorological Proof Packet for PMFBY Claims

PACKET ID: PKT-4

CLAIM DATE: 3/9/2026

STATUS: VERIFIED EVIDENCE

1. FARMER & AGRISTACK REGISTRY DETAILS

Farmer Name:	Ramesh Kumar	AgriStack ID:	AGR-IND-88219
Phone Number:	9876543210	Farmer ID (Local):	FRM-1
Crop Sown:	Wheat	Total Area:	4.34 Hectares

2. SENTINEL-2 L2A SPECTRAL AUDIT (COPERNICUS SATELLITE)

Canopy NDVI:	0.386 (Vegetation Index)	Baseline Expected:	0.500 NDVI
Water Index (NDWI):	-0.377 (Moisture Index)	Vegetation Deficit:	23% Below Norm
Satellite Mission:	Copernicus Sentinel-2 Harmonized	Spatial Resolution:	10m Multi-spectral Pixel

3. METEOROLOGICAL TELEMETRY AUDIT (OPEN-METEO 7-DAY)

Cumulative 7d Rain:	18.90 mm	Drought Threshold:	< 20.0 mm (Deficit)
Avg Max Temp:	34.10 °C	Heat Index Stress:	MODERATE

4. AUTOMATED STRESS DETERMINATION & PMFBY CLAIM ADVICE

STRESS: NONE

Triggered Rule: R0_no_threshold_crossed

Assessed Crop Loss: 35% Estimated Yield Loss Confidence Score: 0% Verified

Evidence Summary: Sensed multi-band reflectance exhibits critical vegetative moisture collapse coinciding with unbroken dry spell and high thermal radiation during reproductive crop phase.

5. CRYPTOGRAPHIC VERIFICATION & AUDIT HASH

SHA-256 EVIDENCE INTEGRITY HASH (TAMPER-EVIDENT):

0x3aedaffbb3772fba1f61af8e395feb762e282062ebaf5798fd6e19fbe1b7e8b7